From O*NET tasks to FEOR-08 task measures

Construction, stability, and a comparison with our own survey

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What I did

- A reproducible pipeline that creates task measures on FEOR-08 (and FEOR-93) from the annual O*NET releases
  - 2019–2025, one file per year, ≈97% of FEOR-08 codes

  Output: `.dta` files at the `FEOR` × `year` level that contain task measures using the two standard definitions (Acemoglu-Autor 2011; Autor-Dorn 2013)

- How stable are the task measures across years?

  At the 1-digit FEOR level they seem stable over 2019–2025.

- Compare with our own survey: same wage regressions on the survey task items and on the O*NET-based measures

		Magnitude of the effect	
Task	Sign of the effect	without firm FE	with firm FE
Abstract	✓	✓	
Routine	✓		✓
Manual	✓		✓

What is O*NET?

Occupational database of the US Department of Labor, successor to the *Dictionary of Occupational Titles (DOT)*.

- Approx. 900 occupations in the O*NET-SOC (Standard Occupational Classification)
- Hundreds of standardized descriptors per occupation, grouped in domains
  - Abilities, Skills, Knowledge
  - Work Activities, Context, Styles . . .
- Scales: Importance or Level, rescaled 0–100, from surveys of job incumbents and from "experts"
  - Work Context items have their own frequency / amount scales
- Standard input of the task-based literature (Autor-Levy-Murnane 2003; Acemoglu-Autor 2011; Autor-Dorn 2013)

Updating:

- 3–4 releases per year; each release re-rates a rotating subset of occupations, and the rest carry over
- I use the latest release of each year

Which O*NET elements make up a task?

The task category → O*NET element mapping is defined once, in a swappable `taskdef_*.do` file.

Two well-known definitions are built:

Acemoglu & Autor (2011) — 5 categories, 16 elements ▸ the 16 elements

- non-routine cognitive: analytical / interpersonal
- routine cognitive / routine manual
- non-routine manual: physical

Autor & Dorn (2013) — 3 aggregates, *same* 16 elements, regrouped ▸ the 16 elements

- abstract = (non-routine cognitive: analytical + interpersonal)
- routine = (routine cognitive + routine manual)
- manual = non-routine manual: physical

From O*NET elements to task composites

1. Standardize every O*NET element **within that year's release**
2. Composite $=$ unweighted mean of the standardized elements in that category
3. Re-standardize the composite $\rightarrow$ `task_*_z`

The crosswalk chain, O*NET-SOC $\rightarrow$ FEOR-08

2020–2025: O*NET-SOC 2019 $\rightarrow$ SOC 2018 $\rightarrow$ SOC 2010 $\rightarrow$ ISCO-08 $\rightarrow$ FEOR-08

2019: O*NET-SOC 2010 (= SOC 2010) $\rightarrow$ ISCO-08 $\rightarrow$ FEOR-08

Crosswalk leg	Source
O*NET database	onetcenter.org, 2019 (24.1) $\rightarrow$ 2025 (30.1)
SOC 2010 $\leftrightarrow$ SOC 2018	BLS
SOC 2010 $\leftrightarrow$ ISCO-08	BLS
ISCO-08 $\rightarrow$ FEOR-08	KSH

Every many-to-one match is collapsed with an **unweighted mean**

Coverage

470 of 485 FEOR-08 four-digit codes covered ($\approx$97%)

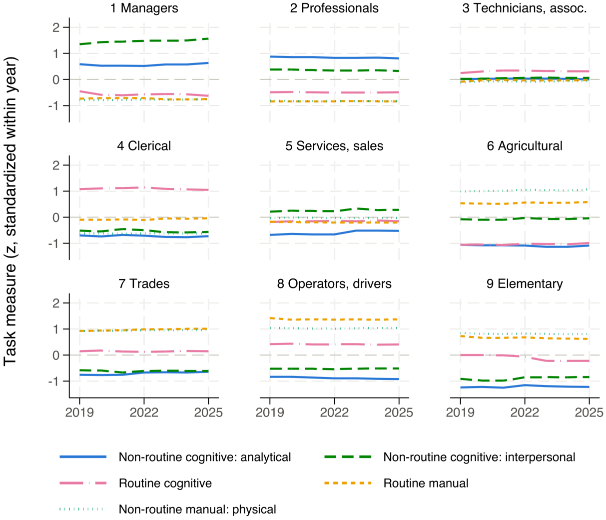
The 15 missing codes are missing for **structural** reasons, not because of pipeline defects:

- **1 code**: no ISCO source in the KSH crosswalk at all (FEOR 3134: Környezetvédelmi technikus)
- **14 codes**: mapped, but trace only to SOC occupations O*NET never rates — military (SOC 55-*), *Legislators* (11-1031), and SOC "All Other" residual codes

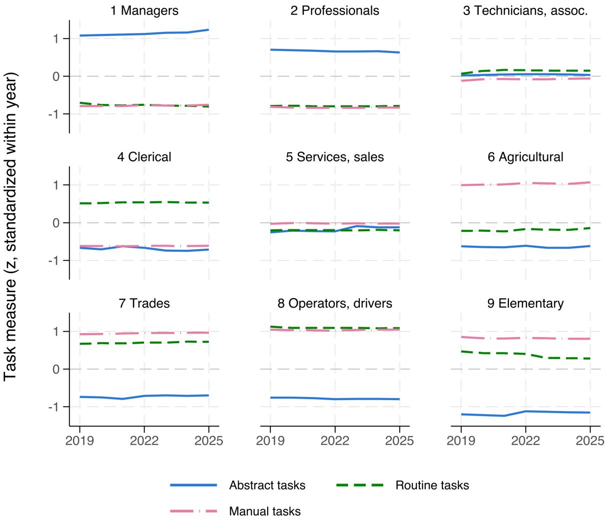
Should FEOR-level task content change over time?

- Composites are standardized **within** each year's O*NET release, so a value is a *relative* position among that year's occupations
- O*NET re-rates only a slice of occupations per release $\Rightarrow$ year-to-year movement in any one occupation's relative position should be small
- **Check**: average the standardized composites within each 1-digit FEOR major group

Trends by 1-digit FEOR: Acemoglu & Autor (2011)



Trends by 1-digit FEOR: Autor & Dorn (2013)



How the comparison is set up

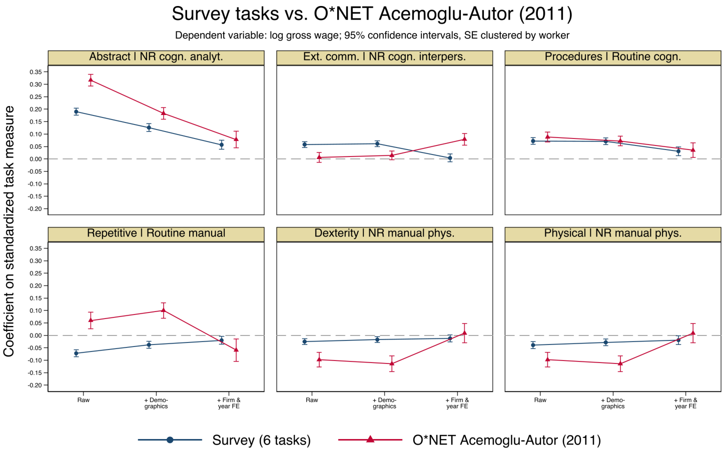
Survey questions aggregated into comparable task measures:

- Abstract: digital use, reading, calculation
- Routine: repetitive work, following procedures
- Manual: physical work, dexterity
- + (External) communication
- Wage regression on survey-based task measures and O*NET-crosswalked counterparts:

$$\ln(\text{wage}_{it}) = \beta\, \text{task}_i + \gamma X_{it} + \delta_j + \theta_t + \varepsilon_{it}$$

  - X_{it}: age, age^2, education, gender, county
  - $\delta_j; \theta_t$: firm and year FE
  - four specifications: Raw $\rightarrow$ + demographics $\rightarrow$ + firm FE $\rightarrow$ + year FE
- Two pairwise comparisons:
  - survey (3 aggregate tasks) vs. **Autor–Dorn (2013)**
  - survey (6 detailed tasks) vs. **Acemoglu–Autor (2011)**

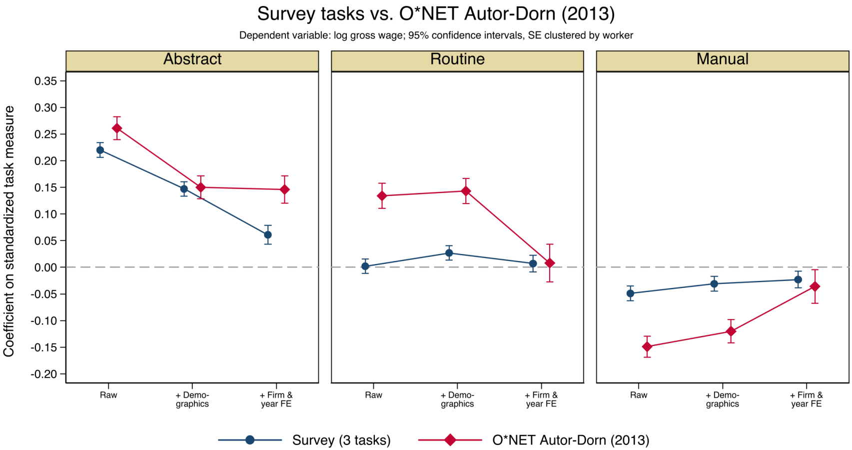
Survey (6 tasks) vs. O*NET Acemoglu–Autor (2011)



$N = 10{,}513$ without firm FE

$N = 5{,}638$ with firm FE

Survey (3 tasks) vs. O*NET Autor–Dorn (2013)



$N = 10{,}513$ without firm FE
$N = 5{,}638$ with firm FE

Wage returns: survey vs. O*NET

	Raw		+ Demographics		+ Firm & year FE	
Task	Survey	O*NET	Survey	O*NET	Survey	O*NET
Survey (3 tasks) vs. AD (2013)						
Abstract	0.220***	0.261***	0.147***	0.150***	0.061***	0.146***
Routine	0.002	0.134***	0.027***	0.143***	0.007	0.008
Manual	−0.049***	−0.149***	−0.031***	−0.120***	−0.023***	−0.036**
*Survey (6 tasks) vs. O*NET Acemoglu–Autor (2011)*						
Abstract	0.190***	0.317***	0.126***	0.183***	0.057***	0.078***
Ext. communication	0.058***	0.006	0.061***	0.014	0.004	0.079***
Follows procedures	0.072***	0.088***	0.071***	0.072***	0.031***	0.035**
Repetitive	−0.072***	0.060***	−0.038***	0.100***	−0.020**	−0.059**
Dexterity	−0.025***	−0.098***	−0.017***	−0.114***	−0.012*	0.009
Physical	−0.039***	−0.098***	−0.028***	−0.114***	−0.019**	0.009
Observations	10,513	10,513	10,513	10,513	5,638	5,638

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$; SE clustered by worker. Dexterity and Physical are both paired to O*NET *non-routine manual: physical*.

Discussion

- Aggregate using employment weights instead of the unweighted means?
- Beyond wages: which other outcomes should we cross-validate on?
- ...

Sources

- Acemoglu, D. & Autor, D. (2011), "Skills, Tasks and Technologies: Implications for Employment and Earnings," *Handbook of Labor Economics* 4B.
- Autor, D. & Dorn, D. (2013), "The Growth of Low-Skill Service Jobs and the Polarization of the US Labor Market," *American Economic Review* 103(5).
- O*NET database, US Department of Labor / Employment and Training Administration, https://www.onetcenter.org
- BLS SOC crosswalks, https://www.bls.gov/soc/
- KSH FEOR-08 documentation, https://www.ksh.hu/feor_menu

Crosswalk code and full documentation:
https://github.com/martinneubrandt/onet-feor

Appendix: the 16 elements, Acemoglu–Autor (2011)

O*NET element	Code
Non-routine cognitive: analytical	
Analyzing data / information	4.A.2.a.4
Thinking creatively	4.A.2.b.2
Interpreting the meaning of information for others	4.A.4.a.1
Non-routine cognitive: interpersonal	
Establishing and maintaining personal relationships	4.A.4.a.4
Guiding, directing and motivating subordinates	4.A.4.b.4
Coaching / developing others	4.A.4.b.5
Routine cognitive	
Importance of repeating the same tasks	4.C.3.b.7
Importance of being exact or accurate	4.C.3.b.4
Structured vs. unstructured work **(reversed)**	4.C.3.b.8

O*NET element	Code
Routine manual	
Pace determined by the speed of equipment	4.C.3.d.3
Controlling machines and processes	4.A.3.a.3
Spend time making repetitive motions	4.C.2.d.1.i
Non-routine manual: physical	
Operating vehicles, mechanized devices or equipment	4.A.3.a.4
Spend time using hands to handle, control or feel objects	4.C.2.d.1.g
Manual dexterity	1.A.2.a.2
Spatial orientation	1.A.1.f.1

Code prefixes: 1.A = Abilities, 4.A = Work Activities, 4.C = Work Context.

Autor–Dorn (2013) regroups the same 16: abstract = the two non-routine cognitive blocks, routine = the two routine blocks, manual = non-routine manual: physical.

Back

Defined in `taskdef_aa2011.do`.

Appendix: the same 16 elements, Autor–Dorn (2013)

O*NET element	Code
Abstract (= NR cognitive: analytical + interpersonal)	
Analyzing data / information	4.A.2.a.4
Thinking creatively	4.A.2.b.2
Interpreting the meaning of information for others	4.A.4.a.1
Establishing and maintaining personal relationships	4.A.4.a.4
Guiding, directing and motivating subordinates	4.A.4.b.4
Coaching / developing others	4.A.4.b.5
Routine (= routine cognitive + routine manual)	
Importance of repeating the same tasks	4.C.3.b.7
Importance of being exact or accurate	4.C.3.b.4
Structured vs. unstructured work **(reversed)**	4.C.3.b.8
Pace determined by the speed of equipment	4.C.3.d.3
Controlling machines and processes	4.A.3.a.3
Spend time making repetitive motions	4.C.2.d.1.i

O*NET element	Code
Manual (= NR manual: physical)	
Operating vehicles, mechanized devices or equipment	4.A.3.a.4
Spend time using hands to handle, control or feel objects	4.C.2.d.1.g
Manual dexterity	1.A.2.a.2
Spatial orientation	1.A.1.f.1

Code prefixes: 1.A = Abilities, 4.A = Work Activities, 4.C = Work Context.

Exactly the elements of the Acemoglu–Autor (2011) definition, only aggregated into three composites instead of five.

Defined in `taskdef_ad2013.do`.

Back

Appendix: pipeline

Step	File	Description
00	`00_master.do`	entry point: runs every step below
01	`01_download_onet.do`	fetch that year's O*NET release
02	`02_build_crosswalks.do`	build the four crosswalk legs
03	`03_append_onet.do`	stack Abilities / Work Activities / Work Context
–	`taskdef_*.do`	define the task definition (elements per category)
04	`04_build_elements.do`	keep only the definition's elements, reshape wide
05	`05_build_measures.do`	standardize, reverse-code, build composites
06	`06_crosswalk_feor.do`	O*NET-SOC $\rightarrow \cdots \rightarrow$ FEOR-08
07	`07_build_panel.do`	stack years 2019–2025 into a panel
08	`08_plot_trends.do`	draw the stability figure

Steps 01–03 and 06–08 are definition-agnostic: a new task definition touches only `taskdef_*.do`. The two BLS files are committed rather than auto-downloaded — BLS returns HTTP 403 to non-browser requests.